MICROPOWERMANAGER • OFFICIAL INTEGRATION

WhatsApp Business API Plugin

Complete Illustrated Configuration, Operations & User Guide for Administrators, Support Operators, Field Agents, and Energy Consumers

PLATFORM
MicroPowerManager (MPM)

PLUGIN VERSION
1.0.0 (Production Release)

GATEWAY TECHNOLOGY
Meta WhatsApp Cloud API (v18.0)

FALLBACK INFRASTRUCTURE
Two-Tier GSM Cellular SMS

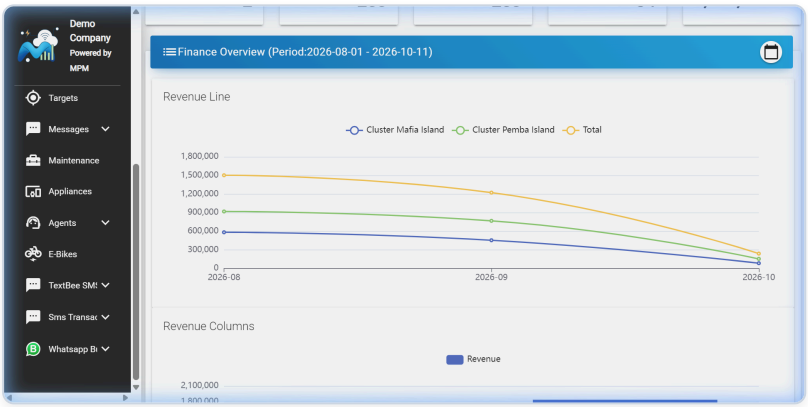


Figure 1.0: MicroPowerManager ERP Dashboard with Integrated WhatsApp Business API Module

Document Scope & Target Stakeholders

- **System Administrators & DevOps:** Meta Cloud API provisioning, credentials, webhooks, and template approvals.
- **Utility Operators & Support Agents:** Unified conversation inbox (`/sms`), opt-in compliance, 24-hour service window, and ticket desks.
- **Field Agents & Cashiers:** Customer registration opt-ins, phone updates, and subscriber orientation.
- **End Consumers & Subscribers:** 24/7 interactive self-service buttons, instant token retrieval, balance checks, and cellular SMS fallback.

1. Executive Summary & System Architecture

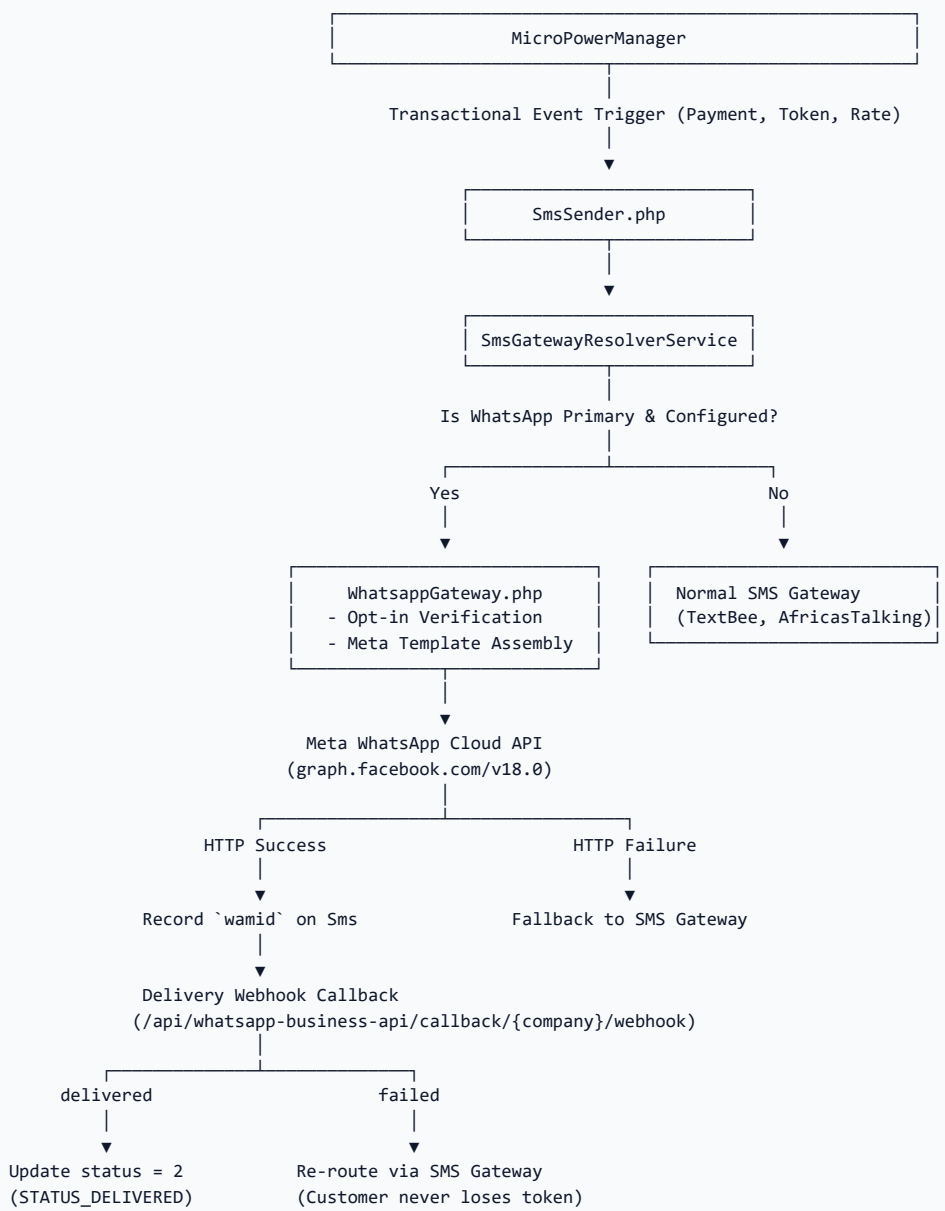
MicroPowerManager (MPM) is an open-source CRM and ERP platform tailored for energy operators, mini-grid utilities, and Solar Home System (SHS) companies operating in Sub-Saharan Africa and South/Southeast Asia. In these markets, WhatsApp is the dominant daily communications channel.

Traditional SMS carries severe commercial and operational limitations for off-grid utilities:

- **High Unit Costs:** Significant portion of operating margins spent on cellular carrier message fees.
- **Silent Failure Rates:** Local GSM networks regularly drop SMS notifications without returning delivery statuses.
- **Customer Distrust:** Consumers struggle to identify alphanumeric shortcodes and unverified numeric senders.

The **WhatsApp Business API Plugin** solves these challenges by routing payment confirmations, electricity token dispatches, and appliance financing reminders directly over the **Meta WhatsApp Cloud API**, while backing every message with an automatic **Two-Tier SMS Fallback** guarantee.

System Communication Flow



Two-Tier Reliability Guarantee: If a customer's smartphone is out of data, switched off, or disconnected from the internet, the plugin automatically detects the delivery timeout via webhook and re-dispatches the electricity token over standard cellular GSM SMS. Customers never lose access to their power.

2. System Administrators & DevOps Configuration Guide

This section details the prerequisites and step-by-step technical procedures required to provision credentials, configure Meta WhatsApp Cloud API, and connect webhooks to MicroPowerManager.

2.1 Prerequisites & System Authentication

1. MicroPowerManager backend running Laravel with public HTTPS domain accessibility.
2. Meta Business Manager account with verified business status (business.facebook.com).
3. Meta Developer Portal account (developers.facebook.com).
4. Dedicated phone number for WhatsApp Business (capable of receiving verification calls/SMS).

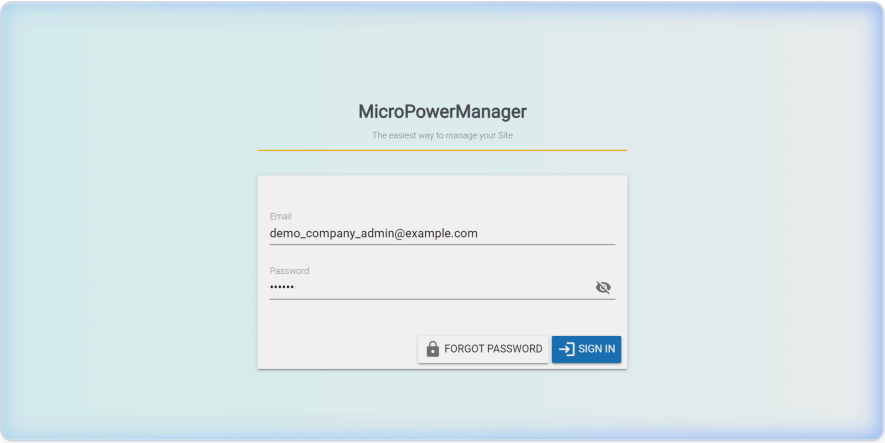


Figure 2.1: MicroPowerManager Administrator Sign-In Interface

2.2 Step 1: Create Meta Cloud API App

1. Log into the Meta for Developers Portal and click **Create App**.
2. Select **Business** as the application type.
3. In the App Dashboard, locate the **WhatsApp** product tile and click **Set up**.
4. Under **WhatsApp** → **API Setup**, copy the following generated identifiers:
   - **Phone Number ID:** The unique identifier for your registered sender number.
   - **WhatsApp Business Account ID (WABA ID):** The ID of your WhatsApp Business Account.

2.3 Step 2: Generate Permanent System User Access Token

> **Never Use Temporary Tokens:** The token provided in the API Setup tab expires in 24 hours. Production deployments must use a permanent token issued to a System User.

1. Navigate to **Meta Business Manager** → **Business Settings** → **Users** → **System Users**.
2. Click **Add** to create a new System User with the `Admin` role (e.g., `mpm-whatsapp-service`).
3. Under **Assigned Assets**, assign your WhatsApp app and grant **Full Control**.
4. Click **Generate New Token**, select your WhatsApp App, and check the following permission scopes:
   - `whatsapp_business_messaging` (required to dispatch and receive messages).
   - `whatsapp_business_management` (required for template retrieval).
5. Save the token securely in your utility's credential store.

2.4 Step 3: Configure Templates in Meta Business Manager

In Meta Business Manager, open **WhatsApp Manager** → **Account Tools** → **Message Templates** and create two templates under category **UTILITY**:

Template Name	Category	Language	Body & Parameter Mappings
`transaction_confirmation`	`UTILITY`	`en`	`Dear customer, your payment of {{1}} has been received. Your electricity token is {{2}}. Thank you for using our service!` • *{{1}} = Amount with currency (e.g., 5000 TZS)* • *{{2}} = Energy Token code (e.g., 4455-6677-8899)*
`appliance_rate_reminder`	`UTILITY`	`en`	`Dear customer, your appliance installment of {{1}} is due on {{2}}. Please make your payment on time to avoid interruption.` • *{{1}} = Remaining balance with currency* • *{{2}} = Due date (e.g., 2026-10-31)*

2.5 Step 4: Configure Plugin Settings in MPM Web UI

1. Log into your MPM dashboard with `Admin` or `Owner` credentials.
2. Navigate to **Settings** → **Plugins** and enable **WhatsApp Business API**.
3. In the left sidebar, click **WhatsApp Business API** → **Overview** (`/whatsapp-business-api/overview`).
4. Enter your credentials into the configuration form:
   - **Phone Number ID:** Meta Phone Number ID.
   - **WhatsApp Business Account ID:** WABA ID.
   - **Access Token:** Permanent System User Token.
   - **Webhook Secret:** A cryptographically strong string (e.g. `WABA_Secret_Token_2026_xyz`).
   - **Transaction Confirmation Template:** `transaction_confirmation` .
   - **Reminder Template:** `appliance_rate_reminder` .
5. Click **Save Credentials** to persist and validate.

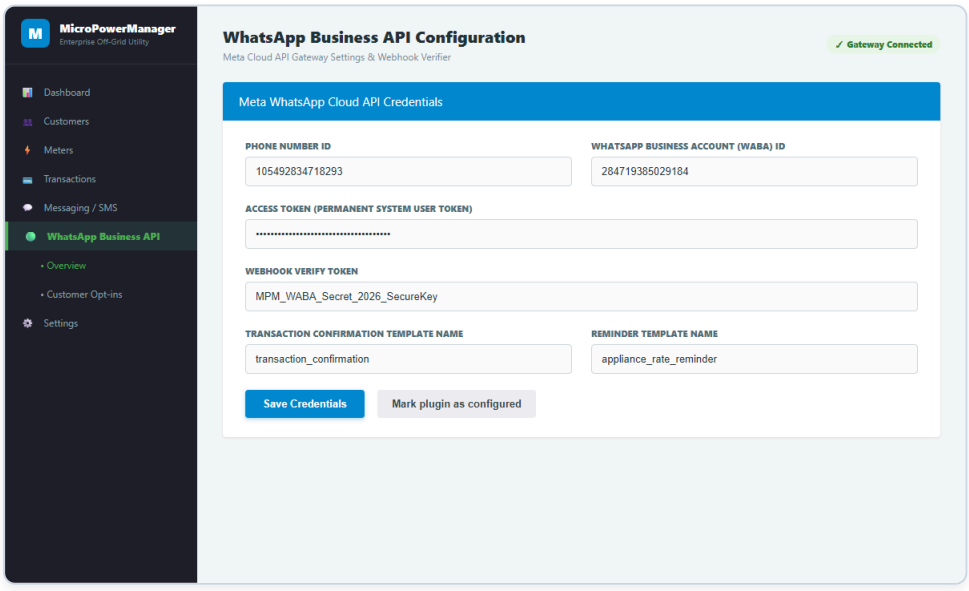


Figure 2.2: WhatsApp Business API Credential Configuration Dashboard (/whatsapp-business-api/overview)

2.6 Step 5: Configure Meta Webhooks

1. Return to the Meta Developer Portal under **WhatsApp → Configuration → Webhook**.
2. Click **Edit** and configure:
   - **Callback URL:** `https://<your-mpm-domain>/api/whatsapp-business-api/callback/<company_id>/webhook`
   - **Verify Token:** The exact **Webhook Secret** configured in MPM.
3. Click **Verify and Save**.
4. Under **Webhook Fields**, click **Manage** and check `messages`.

3. Utility Operators & Customer Support Guide

This section provides operational procedures for customer care teams, call center representatives, and dispatchers managing day-to-day communication.

3.1 Regulatory Opt-In Management

Meta enforces a strict opt-in policy. Businesses may not send unprompted outbound notifications without prior customer consent. The plugin automatically maintains an internal registry of customer consent.

- **Viewing Customer Consent:** Navigate to **WhatsApp Business API** → **Customer Opt-ins** (`/whatsapp-business-api/opt-in`). The list displays every registered customer, their primary mobile number, opt-in status, and last updated timestamp.
- **Toggling Opt-In Status:** Support agents can immediately enable or revoke opt-in for any customer by clicking the toggle switch in the table.
- **Automatic Fallback Enforcement:** If an operator triggers an electricity token or financing reminder to a customer who is **not opted in**, MPM automatically bypasses WhatsApp and sends the notification directly via the secondary SMS gateway.

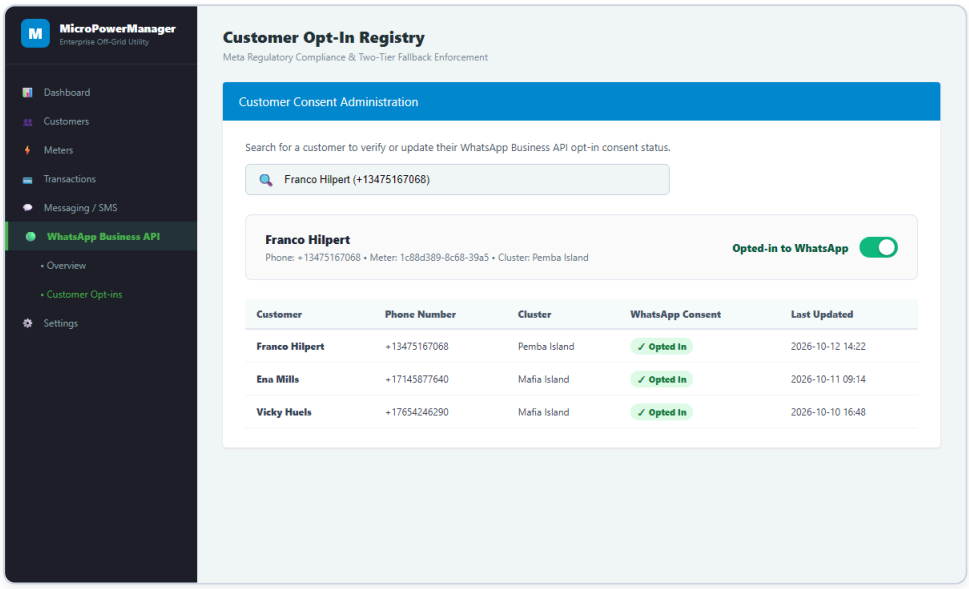


Figure 3.1: Customer Opt-in Registry and Consent Management Interface (/whatsapp-business-api/opt-in)

3.2 Unified Messaging Inbox (`/sms`)

Support agents do not need to switch between different applications to view customer communication. All SMS and WhatsApp interactions are unified in the **Messaging** → **SMS** conversation screen.

Visual Element	Display Behavior	Operator Meaning
WhatsApp Outline Logo	Official WhatsApp Business outline logo with centered 'B' rendered next to message timestamp.	Message was sent or received over the Meta WhatsApp Cloud API.
Standard SMS Icon	Generic `sms` chat bubble icon.	Message was sent or received over cellular GSM (TextBee, Africa's Talking).
Light Blue Bubble (`#e2f3fd`)	Right-aligned outbound bubble.	Automated token delivery, bot response, or manual agent reply.
Soft Red/Pink Bubble (`#ffebee`)	Left-aligned inbound bubble.	Inbound customer inquiry or self-service button click.

3.3 Conversation Thread & Channel Differentiation

In the conversation view below, operators can immediately distinguish between cellular SMS messages and WhatsApp messages through the integrated brand indicators and theme styling:

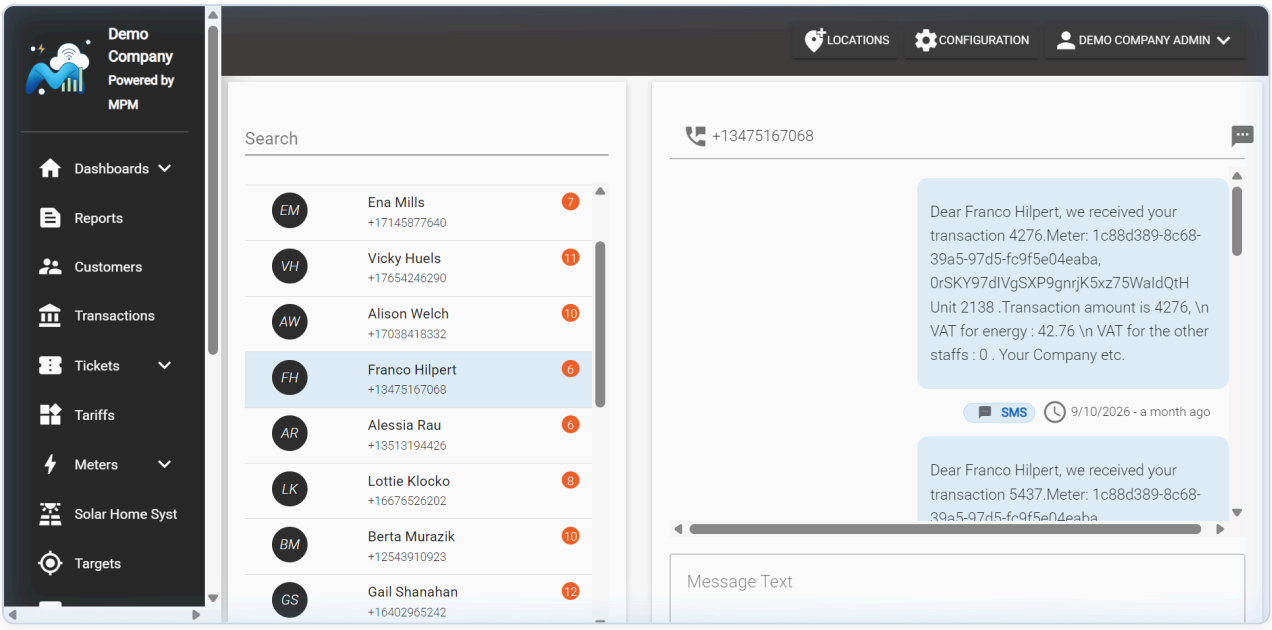


Figure 3.2: Unified Conversation Stream (/sms) displaying real-time customer threads with channel badges

3.4 Replying Within the 24-Hour Customer Care Window

When a customer sends any message to the utility's WhatsApp number, Meta opens a **24-hour Customer Care Window**:

- Within this 24-hour window, support agents can type free-form text in the MPM conversation compose box without being restricted to pre-approved templates.
- If the 24-hour window expires, free-form text messages cannot be delivered over WhatsApp. MPM will automatically route the agent's message via the SMS gateway so the customer still receives it.

3.5 Managing Bot-Escalated Missing Token Tickets

When a customer clicks **"Did not receive token"** and MPM finds no completed transaction on record (e.g. mobile money pending or delayed), the plugin automatically generates an escalated support ticket:

1. Navigate to **Tickets → Open Tickets**.
2. Locate the newly generated ticket titled `Missing Token Report: <Customer Phone>`.
3. The ticket body contains the customer's phone number, meter number, and recent transaction attempt.
4. **Resolution Procedure:**
   - Cross-reference the transaction reference with your mobile money aggregator portal (M-Pesa, Airtel Money, Wave).
   - If payment is confirmed, complete the transaction in MPM to generate the token code.
   - Close the ticket in MPM. The customer is automatically notified.

4. Field Agents & Cashiers Onboarding Guide

Field agents, local retail cashiers, and installation technicians are the primary touchpoint for rural customers. Proper onboarding guarantees high deliverability and customer satisfaction.

4.1 Field Registration Workflow

1. **Verify Active WhatsApp Line:** When signing up a new customer, ask if their mobile phone number is registered on WhatsApp. If they use a different SIM for WhatsApp, enter that number in their primary address record.
2. **Capture Consent Opt-In:** On the MPM customer registration screen, tick the **"Opt in for WhatsApp notifications"** checkbox.
3. **Number Formatting:** Always input phone numbers in full international E.164 format (e.g. `+255712345678` or `+2348012345678`) without leading zeroes or spaces.

4.2 Customer Orientation & Education

> **Best Practice:** Have the customer save the utility's official WhatsApp Business number into their phone's contact list during installation. This ensures that utility messages display the official company logo and verified name rather than an unknown number.

Inform the customer of three key service benefits:

- **Electricity Tokens on WhatsApp:** Tokens arrive on WhatsApp within seconds of completing mobile money payments.
- **Free Self-Service 24/7:** They can text "*hello*" or "*menu*" anytime to retrieve tokens, check balances, or get help.
- **SMS Fallback Assurance:** If they turn off their mobile data or lose internet connection, tokens will automatically arrive via regular SMS.

4.3 Assisting Customers with SIM / Number Changes

If a customer purchases a new SIM card or migrates their WhatsApp account:

- **Option A (Field Agent assisted):** Open the customer's account in MPM and update their phone number in the Address tab. The plugin's `AddressObserver` automatically updates their WhatsApp profile.
- **Option B (Self-service):** Instruct the customer to text `UPDATE NUMBER <NewPhone>` from their WhatsApp application.

5. End-Customer & Consumer User Guide

This section provides a clear, non-technical reference explaining how energy consumers interact with the utility over WhatsApp.

5.1 Automatic Notifications & Self-Service Menu

Consumers can access customer service anytime without calling a call center. Texting any general word (such as *"hi"*, *"hello"*, *"help"*, *"token"*, *"habari"*) immediately opens the interactive service menu:

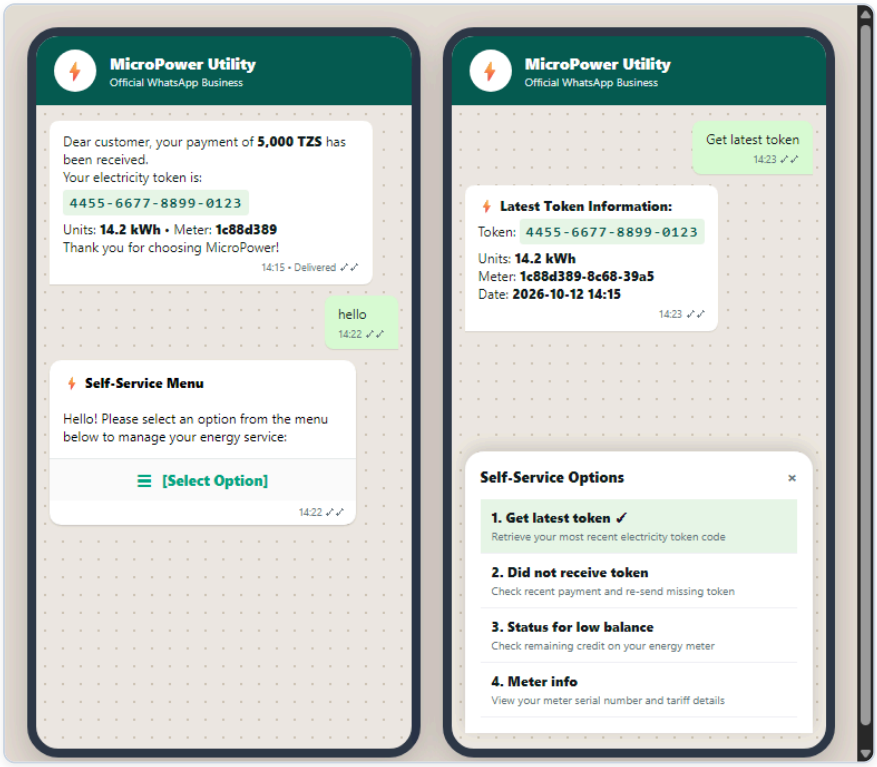


Figure 5.1: Real WhatsApp Consumer Experience • (Left) Token Delivery & Interactive Menu • (Right) 4-Action Selection Sheet & Bot Reply

5.2 Self-Service Options & Natural Text Commands

Customers can either tap the buttons in the WhatsApp sheet or type the action number or natural phrase:

Action	How Customer Triggers It	What Customer Receives
Get Latest Token	Tap `Get latest token`, type `1`, or text *"latest token"* or *"to get latest token"*	Most recent electricity token code (e.g. `4455-6677-8899`), units (`15.4 kWh`), meter serial number, and timestamp.
Did Not Receive Token	Tap `Did not receive token`, type `2`, or text *"did not recieve token"* or *"no token"*	Immediate re-send of completed token, or ticket confirmation (`#123`) with prompt escalation to support agents.
Check Balance	Tap `Status for low balance`, type `3`, or text *"status for low balance"* or *"balance"*	Current credit on meter (`kWh`). If balance is ≤ 5.0 kWh, warns of low balance and recommends recharging.
Meter Information	Tap `Meter info`, type `4`, or text *"meter info"* or *"my meter"*	Registered meter serial number, tariff category, connection phase, and meter status.
Update Phone Number	Text `UPDATE NUMBER +255...` or `BADILI NAMBA +255...`	Automated confirmation and instant account synchronization with the new phone number.

6. Troubleshooting & Diagnostic Matrix

Use this diagnostic reference to quickly identify and resolve operational and technical anomalies.

Issue Description	Probable Root Cause	Resolution Procedure
Webhook verification fails with HTTP 403	Verify token mismatch between Meta Developer Portal and MPM.	Open **WhatsApp Business API** → **Overview** in MPM, copy the **Webhook Secret**, and paste it into Meta's **Verify Token** field. Save changes.
Outbound token sent via SMS instead of WhatsApp	Customer's `opted_in` flag is false or unrecorded.	Navigate to **WhatsApp Business API** → **Customer Opt-ins**, search for the customer, and toggle their opt-in switch to **Active**.
Meta API error: `Template not found`	Template name in MPM Overview does not exactly match Meta Business Manager.	Check the approved template names in Meta Business Manager. Update the template name in MPM Overview to match exactly (case-sensitive).
Customer messages do not trigger the self-service menu	Meta webhook field subscription for `messages` is missing or disabled.	In Meta Developer Portal, go to **WhatsApp** → **Configuration** → **Webhook Fields**, click **Manage**, and ensure `messages` is checked.
Customer receives no WhatsApp message when phone is offline	Normal behavior. Recipient has no mobile data or Wi-Fi.	No manual intervention required. When Meta reports delivery timeout/failure, MPM's two-tier fallback automatically dispatches the message via cellular SMS.
Free-form support reply fails to send	The 24-hour customer care window has lapsed since customer's last message.	Meta restricts free-form messages after 24 hours. The plugin will automatically dispatch the agent's message via SMS gateway instead.

Site Administrator Health-Check Checklist

- ☑ **Meta App Status:** App mode is set to **Live** (not Development).
- ☑ **System User Token:** Token is permanent with `whatsapp_business_messaging` scope.
- ☑ **Payment Method in Meta:** Verified payment method attached to Meta Business Account for messaging quotas.
- ☑ **Webhook HTTPS:** MPM domain serves a valid, trusted SSL/TLS certificate (Let's Encrypt or commercial CA).
- ☑ **Queue Worker Daemon:** Laravel queue worker (`php artisan queue:work`) is active and supervised.
- ☑ **Secondary SMS Gateway:** Fallback SMS gateway (TextBee / Africa's Talking) has active credits.

© 2026 MicroPowerManager Contributors • Open-Source Distributed Utility Management System